	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1\,\alpha\beta\chi}$	$\mathcal{K}_{1^+}^{\#1\,\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2\,\alpha\beta}$	$\mathcal{K}_{1^-}^{\#1\,\alpha}$	$\mathcal{K}_{1^-}^{\#2\,\alpha}$	$\mathcal{K}_{2^+}^{\#1\,\alpha\beta}$	$\mathcal{K}_{2^-}^{\#1\,\alpha\beta\chi}$
$\mathcal{K}_{0^+}^{\#1\,\dagger}$	0	0						
$\mathcal{K}_{0^-}^{\#1\,\dagger\alpha\beta\chi}$	0	0						
$\mathcal{K}_{1^+}^{\#1\,\dagger\alpha\beta}$			$-\frac{Y_1 k^2}{2}$	$\frac{k^2\,{}^{(4)}\kappa_4}{2\sqrt{2}}$	0	0		
$\mathcal{K}_{1^+}^{\#2\,\dagger\alpha\beta}$			$\frac{k^2\,{}^{(4)}\kappa_4}{2\sqrt{2}}$	$\frac{Y_2 k^2}{2}$	0	0		
$\mathcal{K}_{1^-}^{\#1\,\dagger\alpha}$			0	0	0	0		
$\mathcal{K}_{1^-}^{\#2\,\dagger\alpha}$			0	0	0	0		
$\mathcal{K}_{2^+}^{\#1\,\dagger\alpha\beta}$							$\frac{3k^2\,{}^{(4)}\kappa_1}{2}$	0
$\mathcal{K}_{2^-}^{\#1\,\dagger\alpha\beta\chi}$							0	0

	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^-}^{\#1\,\alpha\beta\chi}$	$\mathcal{J}_{1^+}^{\#1\,\alpha\beta}$	$\mathcal{J}_{1^+}^{\#2\,\alpha\beta}$	$\mathcal{J}_{1^-}^{\#1\,\alpha}$	$\mathcal{J}_{1^-}^{\#2\,\alpha}$	$\mathcal{J}_{2^+}^{\#1\,\alpha\beta}$	$\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi}$
$\mathcal{J}_{0^+}^{\#1\,\dagger}$	0	0						
$\mathcal{J}_{0^-}^{\#1\,\dagger\alpha\beta\chi}$	0	0						
$\mathcal{J}_{1^+}^{\#1\,\dagger\alpha\beta}$			$\frac{Y_2 k^2}{2\,\mathrm{Det}(1^+)}$	$-\frac{k^2\,{}^{(4)}\kappa_4}{2\sqrt{2}\,\mathrm{Det}(1^+)}$	0	0		
$\mathcal{J}_{1^+}^{\#2\,\dagger\alpha\beta}$			$-\frac{k^2\,{}^{(4)}\kappa_4}{2\sqrt{2}\,\mathrm{Det}(1^+)}$	$-\frac{Y_1 k^2}{2\,\mathrm{Det}(1^+)}$	0	0		
$\mathcal{J}_{1^-}^{\#1\,\dagger\alpha}$			0	0	0	0		
$\mathcal{J}_{1^-}^{\#2\,\dagger\alpha}$			0	0	0	0		
$\mathcal{J}_{2^+}^{\#1\,\dagger\alpha\beta}$							$\frac{2}{3k^2\,{}^{(4)}\kappa_1}$	0
$\mathcal{J}_{2^-}^{\#1\,\dagger\alpha\beta\chi}$							0	0

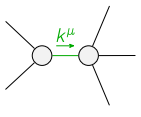
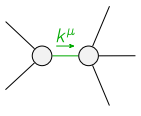
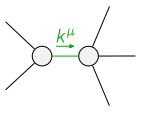
Abbreviations used in matrices

$$Y_1 == 3\,{}^{(4)}\kappa_1 + 2\,{}^{(4)}\kappa_3 \;\&\&\; Y_2 == {}^{(4)}\kappa_3 + {}^{(4)}\kappa_4 \;\&\&\; \mathrm{Det}(1^+) == -\frac{1}{8}k^4\,(6\,{}^{(4)}\kappa_1\,({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + (2\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2)$$

Lagrangian

$$\begin{aligned}&{}^{(4)}\kappa_1\,\partial_\beta\mathcal{K}^{\delta}{}_{\chi\delta}\,\partial^\chi\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + {}^{(4)}\kappa_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\alpha\chi}{}^{\delta} + {}^{(4)}\kappa_4\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\beta\chi}{}^{\delta} -\\ &3\,{}^{(4)}\kappa_1\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} - {}^{(4)}\kappa_3\,\partial_\beta\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}_{\chi\alpha}{}^{\delta} + \frac{1}{2}\,{}^{(4)}\kappa_3\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{2}\,{}^{(4)}\kappa_4\,\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\,\partial_\delta\mathcal{K}^{\delta}{}_{\beta\chi}\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\;\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{0^-}^{\#1\,\alpha\beta\chi} == 0$	1	$\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + \partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + \partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + \partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + \partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + \partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + \partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + \partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + \partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta\mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$
$\mathcal{J}_{1^-}^{\#1\,\alpha} == 0$	3	$\partial_\chi\partial^\alpha\mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi\partial^\chi\mathcal{J}^{\beta\alpha}{}_{\beta} == \partial_\chi\partial_\beta\mathcal{J}^{\beta\alpha\chi}$
$\mathcal{J}_{2^-}^{\#1\,\alpha\beta\chi} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\alpha}{}_{\delta} == 3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon\mathcal{J}^{\delta\beta}{}_{\delta}$
Total # constraint(s):	10	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{1}{{}^{(4)}\kappa_1\,(6\,{}^{(4)}\kappa_1\,({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + (2\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2)}$ Root[15 552 ${}^{(4)}\kappa_1{}^4\,{}^{(4)}\kappa_3{}^2$ + 20 736 ${}^{(4)}\kappa_1{}^3\,{}^{(4)}\kappa_3{}^3$ + 6912 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_3{}^4$ + 31 104 ${}^{(4)}\kappa_1{}^4\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4$ + 41 472 ${}^{(4)}\kappa_1{}^3\,{}^{(4)}\kappa_3{}^2\,{}^{(4)}\kappa_4$ + 13 824 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_3{}^3\,{}^{(4)}\kappa_4$ + 15 552 ${}^{(4)}\kappa_1{}^4\,{}^{(4)}\kappa_4{}^2$ + 25 920 ${}^{(4)}\kappa_1{}^3\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4{}^2$ + 10 368 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_3{}^2\,{}^{(4)}\kappa_4{}^2$ + 5184 ${}^{(4)}\kappa_1{}^3\,{}^{(4)}\kappa_3{}^3$ + 3456 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4{}^3$ + 432 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_4{}^4$ + (432 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_3{}^2$ + 288 ${}^{(4)}\kappa_1\,{}^{(4)}\kappa_3{}^3$ - 432 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_4{}^2$ - 216 ${}^{(4)}\kappa_1\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4{}^2$ - 72 ${}^{(4)}\kappa_1\,{}^{(4)}\kappa_4{}^3$) #1 + (-72 ${}^{(4)}\kappa_1{}^2$ - 48 ${}^{(4)}\kappa_1\,{}^{(4)}\kappa_3$ - 12 ${}^{(4)}\kappa_3{}^2$ + 12 ${}^{(4)}\kappa_1\,{}^{(4)}\kappa_4$ - 12 ${}^{(4)}\kappa_3\,{}^{(4)}\kappa_4$ - 3 ${}^{(4)}\kappa_4{}^2$) #1^2 + #1^3 &, 1]
	2	0	$\frac{1}{{}^{(4)}\kappa_1\,(6\,{}^{(4)}\kappa_1\,({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + (2\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2)}$ Root[15 552 ${}^{(4)}\kappa_1{}^4\,{}^{(4)}\kappa_3{}^2$ + 20 736 ${}^{(4)}\kappa_1{}^3\,{}^{(4)}\kappa_3{}^3$ + 6912 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_3{}^4$ + 31 104 ${}^{(4)}\kappa_1{}^4\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4$ + 41 472 ${}^{(4)}\kappa_1{}^3\,{}^{(4)}\kappa_3{}^2\,{}^{(4)}\kappa_4$ + 13 824 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_3{}^3\,{}^{(4)}\kappa_4$ + 15 552 ${}^{(4)}\kappa_1{}^4\,{}^{(4)}\kappa_4{}^2$ + 25 920 ${}^{(4)}\kappa_1{}^3\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4{}^2$ + 10 368 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_3{}^2\,{}^{(4)}\kappa_4{}^2$ + 5184 ${}^{(4)}\kappa_1{}^3\,{}^{(4)}\kappa_3{}^3$ + 3456 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4{}^3$ + 432 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_4{}^4$ + (432 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_3{}^2$ + 288 ${}^{(4)}\kappa_1\,{}^{(4)}\kappa_3{}^3$ - 432 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_4{}^2$ - 216 ${}^{(4)}\kappa_1\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4{}^2$ - 72 ${}^{(4)}\kappa_1\,{}^{(4)}\kappa_4{}^3$) #1 + (-72 ${}^{(4)}\kappa_1{}^2$ - 48 ${}^{(4)}\kappa_1\,{}^{(4)}\kappa_3$ - 12 ${}^{(4)}\kappa_3{}^2$ + 12 ${}^{(4)}\kappa_1\,{}^{(4)}\kappa_4$ - 12 ${}^{(4)}\kappa_3\,{}^{(4)}\kappa_4$ - 3 ${}^{(4)}\kappa_4{}^2$) #1^2 + #1^3 &, 2]
	2	0	$\frac{1}{{}^{(4)}\kappa_1\,(6\,{}^{(4)}\kappa_1\,({}^{(4)}\kappa_3 + {}^{(4)}\kappa_4) + (2\,{}^{(4)}\kappa_3 + {}^{(4)}\kappa_4)^2)}$ Root[15 552 ${}^{(4)}\kappa_1{}^4\,{}^{(4)}\kappa_3{}^2$ + 20 736 ${}^{(4)}\kappa_1{}^3\,{}^{(4)}\kappa_3{}^3$ + 6912 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_3{}^4$ + 31 104 ${}^{(4)}\kappa_1{}^4\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4$ + 41 472 ${}^{(4)}\kappa_1{}^3\,{}^{(4)}\kappa_3{}^2\,{}^{(4)}\kappa_4$ + 13 824 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_3{}^3\,{}^{(4)}\kappa_4$ + 15 552 ${}^{(4)}\kappa_1{}^4\,{}^{(4)}\kappa_4{}^2$ + 25 920 ${}^{(4)}\kappa_1{}^3\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4{}^2$ + 10 368 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_3{}^2\,{}^{(4)}\kappa_4{}^2$ + 5184 ${}^{(4)}\kappa_1{}^3\,{}^{(4)}\kappa_3{}^3$ + 3456 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4{}^3$ + 432 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_4{}^4$ + (432 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_3{}^2$ + 288 ${}^{(4)}\kappa_1\,{}^{(4)}\kappa_3{}^3$ - 432 ${}^{(4)}\kappa_1{}^2\,{}^{(4)}\kappa_4{}^2$ - 216 ${}^{(4)}\kappa_1\,{}^{(4)}\kappa_3\,{}^{(4)}\kappa_4{}^2$ - 72 ${}^{(4)}\kappa_1\,{}^{(4)}\kappa_4{}^3$) #1 + (-72 ${}^{(4)}\kappa_1{}^2$ - 48 ${}^{(4)}\kappa_1\,{}^{(4)}\kappa_3$ - 12 ${}^{(4)}\kappa_3{}^2$ + 12 ${}^{(4)}\kappa_1\,{}^{(4)}\kappa_4$ - 12 ${}^{(4)}\kappa_3\,{}^{(4)}\kappa_4$ - 3 ${}^{(4)}\kappa_4{}^2$) #1^2 + #1^3 &, 3]
Resolved unitarity condition(s):		(Demonstrably impossible)	